train input, full runs — small-fa

L1D 1 sets x 64, L2 1 sets x 512 · metric schema 9 · gem5 TimingSimpleCPU · 4 November 2026

31 of 32 runs completed, 1 running, 0 pending; **1 remaining**.

Program (does not depend on the cache)

run	configs done	spill/ROI	spill ld/load	spill st/store	(spill+csave)/ROI	(spill+csave) ld/load	(spill+csave) st/store
cactuBSSN	1/1	21.33%	51.32%	55.49%	21.43%	51.50%	55.81%
parest	1/1	0.18%	0.45%	1.28%	0.49%	0.93%	7.38%
deepsjeng	1/1	2.99%	9.41%	15.98%	8.19%	23.80%	48.29%
leela	1/1	0.43%	1.55%	2.04%	6.08%	15.99%	46.72%
namd	1/1	3.22%	8.11%	17.74%	3.23%	8.13%	17.81%
lbm	1/1	9.69%	27.97%	35.99%	9.69%	27.97%	35.99%
nab	1/1	7.61%	25.91%	37.53%	8.29%	27.38%	44.95%
nab_gcn4dna	1/1	7.30%	24.89%	35.20%	8.25%	26.98%	45.65%
xz	1/1	1.36%	4.25%	4.82%	6.14%	16.52%	26.27%
xz_combined	1/1	2.79%	10.48%	25.95%	4.09%	14.43%	41.80%
imagick	1/1	0.04%	0.21%	4.19%	0.25%	0.82%	48.78%
blender	1/1	0.31%	0.55%	4.03%	3.18%	5.93%	39.04%
povray	1/1	0.27%	0.46%	0.95%	14.26%	22.21%	55.40%
omnetpp	1/1	0.01%	0.02%	0.03%	12.59%	24.64%	48.38%
xalancbmk	1/1	0.44%	1.00%	1.04%	18.29%	32.42%	63.05%
x264	1/1	6.30%	13.42%	40.78%	8.75%	18.41%	57.31%
gcc_200	1/1	1.16%	2.64%	4.34%	16.68%	35.31%	67.49%
gcc_scilab	1/1	1.25%	2.90%	4.70%	15.74%	33.65%	65.08%
gcc	1/1	1.41%	3.09%	5.21%	16.29%	32.84%	65.64%
mcf	1/1	1.61%	2.27%	13.28%	2.75%	4.04%	21.88%
perlbench_scrabbl	1/1	0.23%	0.30%	0.93%	17.65%	32.90%	56.14%
perlbench_suns	1/1	0.98%	2.06%	2.86%	18.85%	36.45%	59.85%
perlbench_diffmail	1/1	2.06%	4.51%	4.76%	17.54%	31.15%	53.97%
perlbench_perfect	1/1	1.44%	2.34%	5.00%	18.85%	33.31%	60.43%
perlbench_splitmail	1/1	5.79%	9.36%	20.68%	17.56%	30.55%	58.86%
bwaves_1	1/1	2.99%	7.65%	18.86%	2.99%	7.65%	18.86%
bwaves_2	1/1	3.04%	7.82%	18.83%	3.04%	7.82%	18.83%
exchange2	1/1	4.84%	10.67%	12.41%	6.03%	13.09%	15.72%
fotonik3d	1/1	3.99%	10.66%	12.65%	4.00%	10.69%	12.75%
roms	0/1	—	—	—	—	—	—
wrf	1/1	0.92%	3.44%	2.61%	1.59%	4.97%	8.09%
cam4	1/1	2.49%	9.15%	7.96%	4.59%	14.65%	19.40%
pooled		3.60%	9.44%	17.30%	7.34%	17.15%	41.91%
mean		3.18%	8.35%	13.49%	9.59%	20.87%	41.47%

Miss rates

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
cactuBSSN	done	15.23%	4.85%	33.96%	5.15%	12.45%	59.95%	12.41%	2.86%
parest	done	9.87%	1.06%	10.56%	1.25%	1.68%	72.07%	5.55%	14.15%
deepsjeng	done	7.79%	1.45%	9.88%	7.51%	4.35%	6.05%	2.15%	10.50%

run	state	L1D	spill ld L1	other ld L1	spill st L1	other st L1	L2	spill ld L2	spill st L2
leela	done	4.49%	0.20%	5.23%	0.20%	2.49%	22.42%	0.04%	98.69%
namd	done	11.49%	2.42%	14.61%	1.26%	2.43%	27.80%	7.69%	15.51%
lbm	done	20.86%	0.00%	7.41%	0.37%	72.77%	99.62%	—	29.21%
nab	done	3.87%	0.09%	6.11%	0.17%	0.57%	7.20%	0.52%	1.97%
nab_gcn4dna	done	3.91%	0.06%	6.13%	0.21%	0.46%	74.36%	24.66%	4.31%
xz	done	4.77%	0.03%	7.48%	0.70%	0.57%	47.29%	0.00%	10.38%
xz_combined	done	9.79%	0.84%	12.72%	0.51%	3.88%	22.72%	0.01%	2.94%
imagick	done	5.01%	34.95%	5.00%	0.40%	1.76%	94.98%	1.01%	6.14%
blender	done	6.70%	4.77%	7.32%	2.35%	2.78%	11.68%	1.99%	18.36%
povray	done	8.60%	7.07%	10.48%	19.32%	3.80%	51.48%	71.97%	68.16%
omnetpp	done	16.03%	0.05%	21.36%	0.79%	5.55%	28.39%	23.39%	1.45%
xalancbmk	done	11.44%	6.74%	14.75%	32.17%	4.80%	6.67%	0.00%	0.01%
x264	done	1.80%	0.15%	2.20%	0.83%	1.81%	33.73%	2.63%	16.31%
gcc_200	done	7.53%	0.81%	10.03%	4.07%	3.14%	43.03%	4.22%	7.80%
gcc_scilab	done	8.13%	1.01%	10.86%	3.83%	3.32%	42.90%	2.93%	9.17%
gcc	done	7.80%	0.74%	10.38%	4.11%	3.12%	46.27%	5.37%	7.98%
mcf	done	22.96%	3.67%	26.70%	7.07%	7.37%	77.43%	7.74%	17.84%
perlbench_scrabbl	done	4.94%	5.08%	7.05%	5.74%	1.33%	6.46%	0.00%	0.02%
perlbench_suns	done	4.62%	0.13%	6.60%	3.31%	1.50%	20.06%	0.67%	11.88%
perlbench_diffmail	done	3.03%	0.15%	4.27%	2.24%	1.03%	53.90%	0.03%	24.65%
perlbench_perfect	done	7.52%	0.45%	10.69%	5.10%	2.11%	56.72%	58.32%	31.08%
perlbench_splitmail	done	3.49%	0.05%	4.99%	1.87%	1.56%	20.52%	1.47%	2.58%
bwaves_1	done	7.29%	0.74%	7.71%	0.00%	10.06%	97.31%	14.74%	98.73%
bwaves_2	done	7.29%	0.74%	7.71%	0.00%	10.13%	97.23%	14.39%	98.71%
exchange2	done	1.32%	1.29%	1.84%	1.88%	0.52%	2.51%	0.31%	1.23%
fotonik3d	done	8.39%	1.99%	11.57%	0.01%	0.12%	82.20%	2.28%	24.30%
roms	running	—	—	—	—	—	—	—	—
wrf	done	5.78%	5.29%	5.76%	4.18%	5.97%	61.40%	10.19%	18.33%
cam4	done	9.81%	1.48%	10.17%	1.85%	11.47%	60.40%	12.18%	37.25%
pooled		8.18%	2.52%	9.85%	3.36%	5.42%	46.91%	10.69%	6.05%
mean		8.11%	2.85%	10.05%	3.82%	5.96%	46.28%	9.63%	22.34%

Formulas

column	formula	aciklama
spill/ROI	`((spill_load + spill_store) / roi_total_insts) * 100`	ROI komutlarinin yuzde kaci spill load veya spill store
spill ld/load	`(spill_load / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload
spill st/store	`(spill_store / roi_store) * 100`	ROI store'larinin yuzde kaci spill store
(spill+csave)/ROI	`((spill_load + spill_store + csave_load + csave_store) / roi_total_insts) * 100`	ayni oran, callee-save save/restore dahil
(spill+csave) ld/load	`((spill_load + csave_load) / roi_load) * 100`	ROI load'larinin yuzde kaci spill reload veya callee-save restore
(spill+csave) st/store	`((spill_store + csave_store) / roi_store) * 100`	ROI store'larinin yuzde kaci spill store veya callee-save save
L1D	`(L1D misses / L1D accesses) * 100`	Tum load ve store access'lerinin (spill + other) L1D miss rate'i: L1D'nin bu konfigurasyondaki genel performansi.
spill ld L1	`(spill load L1D misses / spill load L1D accesses) * 100`	Spill load'larin yuzde kaci L1D'de miss oldu. Genelde dusuktur: spill slot'u kisa sure once yazildigi icin cache line cogu zaman hala L1D'dedir.
other ld L1	`((load misses - spill load misses) / (load accesses - spill load accesses)) * 100`	Other load'larin (spill olmayan load'lar) L1D miss rate'i. spill_ld_L1miss ile yan yana bakilir: spill load'lar normal veriden daha mi az, daha mi cok miss oluyor?

column	formula	aciklama
spill st L1	`(spill store L1D misses / spill store L1D accesses) * 100`	Spill store'larin yuzde kaci L1D'de miss oldu. Cache write-allocate: miss olan bir store once cache line'i L2'den getirir, sonra yazar.
other st L1	`((store misses - spill store misses) / (store accesses - spill store accesses)) * 100`	Other store'larin (spill olmayan store'lar) L1D miss rate'i; spill_st_L1miss ile karsilastirmak icin.
L2	`(L2 misses / L2 accesses) * 100`	L2'ye gelen tum isteklerin (L1D ve L1I miss'leri) yuzde kaci L2'de de miss oldu ve DRAM'e gitti.
spill ld L2	`(spill load L2 misses / spill load L2 accesses) * 100`	L1D'de miss olan spill load'larin yuzde kaci L2'de de miss oldu. Payda sadece L1D'yi miss eden spill load'lardir, butun spill load'lar degil.
spill st L2	`(spill store L2 misses / spill store L2 accesses) * 100`	L1D'de miss olan spill store'larin yuzde kaci L2'de de miss oldu (payda yine sadece L1D miss'leri).

spill = compiler-confirmed register-allocator spill (GCC `-dP`), csave = callee-saved register save/restore (CFI), both inside the ROI only. pooled = sum of numerators / sum of denominators over the completed runs; mean = average of the per-run percentages. Miss rates are per cache access; an L2 rate's denominator is the L1D misses of that access type.